

Lecture Notes on Agentic AI

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4 From Base Model to Specialized Tool

The previous lecture explained how a model represents and understands language. This lecture asks a different question: a freshly pre-trained model can predict plausible next tokens, but how does it become the specific, well-behaved tool you actually interact with? Learning objectives:

1. Explain fine-tuning as specialization of a base model
2. Explain RLHF as a way of aligning model behavior with human preference
3. Explain Chain-of-Thought prompting and why it improves reasoning
4. Distinguish reasoning models from standard LLMs
5. Explain multimodal models and Small Language Models (SLMs)
6. Explain distillation and quantization as two different ways of making models cheaper to run

4.1 Fine-Tuning

Definition

Fine-tuning is the process of further training a base model on a specific set of examples (often question-answer pairs) to adapt it for a **specialized task**.

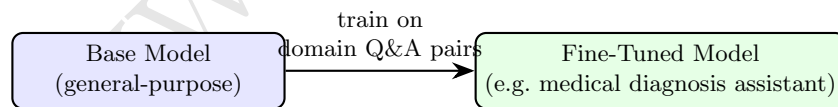


Figure 1: Fine-tuning adapts a general-purpose base model into a specialized tool by further training it on task-specific examples.

Example

Customer support fine-tuning. A base model knows language in general but has no notion of *your* company's refund policy. Fine-tuning it on thousands of real support tickets and correct resolutions teaches it your company's specific procedures, tone, and edge cases.

4.2 RLHF: Reinforcement Learning from Human Feedback

Fine-tuning on question-answer pairs teaches a model *what* to say. RLHF is a different, complementary technique that teaches a model *which of several plausible answers humans actually prefer*.

Definition

RLHF uses **human feedback** to reward response patterns people prefer and penalize those they do not, adjusting the model's future behavior accordingly.

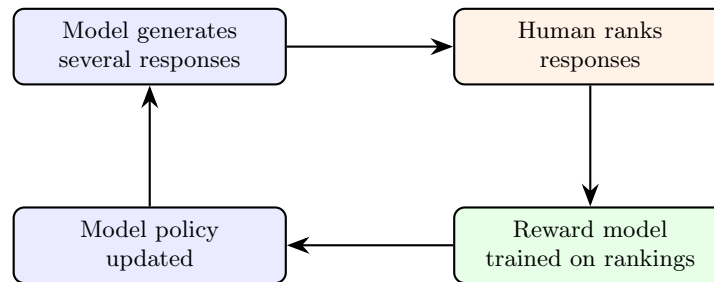


Figure 2: The RLHF loop: the model's own outputs, ranked by humans, are used to train a reward signal that reshapes the model's future behavior. The full mechanics of this loop are covered in a later lecture.

Question

Is RLHF the same as fine-tuning? They are related but distinct. Fine-tuning (Section 4.1) typically trains on fixed, correct question-answer pairs. RLHF trains on *preferences between the model's own generated outputs* – it is less about teaching new facts and more about shaping style, safety, and helpfulness.

4.3 Chain of Thought (CoT)

Definition

Chain of Thought is a technique – used in both prompting and training – where a model breaks a problem down **step by step** rather than jumping straight to a final answer.

Example

Direct answer vs. Chain of Thought. Consider: “A store has 23 apples. It sells 8 and receives a shipment of 15 more. How many apples does it have?”

- *Direct answer:* “30.” (Sometimes right, often wrong – no way to check the reasoning.)
- *Chain of Thought:* “Start with 23. Sell 8: $23 - 8 = 15$. Receive 15 more: $15 + 15 = 30$. Answer: 30.”

The second version is far more likely to be correct on harder problems, and – just as importantly – a human can check *where* it went wrong if it makes a mistake.

4.4 Reasoning Models

Chain of Thought can be *prompted* on an ordinary model. Some models are instead specifically *trained* to reason at length before answering.

Definition

Reasoning models are models trained to determine, on their own, **how many reasoning steps a problem requires**, spending proportionally more computation on harder problems before producing a final answer.

Example

Examples. OpenAI’s o-series and DeepSeek’s R1 are reasoning models: given a hard math or coding problem, they will generate a long internal reasoning trace before answering, whereas a standard chat model answers almost immediately. We revisit this idea, and why it works, in a later lecture on test-time compute.

4.5 Multimodal Models**Definition**

Multimodal models are trained on more than one type of data – text, images, video, or audio – rather than text alone.

Example

Why multimodal helps. A text-only model only knows what an “apple” is from how the word is used in writing. A multimodal model has also seen millions of images labeled “apple,” giving it a grounded, visual understanding of the object itself, not just the word.

4.6 Small Language Models (SLMs)

Not every task needs a trillion-parameter model.

Definition

Small Language Models are task-specific models with far fewer parameters than an LLM (roughly 3 million to 300 million, compared to tens or hundreds of billions), making them cheaper and faster to run.

Property	LLM	SLM
Parameters	Billions – trillions	~3M – 300M
Scope	General-purpose	Task-specific
Hosting cost	High	Low
Typical use	Open-ended reasoning	Narrow, repeated tasks

Table 1: LLM vs. SLM.

4.7 Distillation

A natural question follows: can we get some of an LLM’s quality into a much smaller, cheaper SLM?

Definition

Distillation trains a small “student” model to **mimic the outputs** of a larger “teacher” model, transferring much of the teacher’s quality into a far smaller architecture.

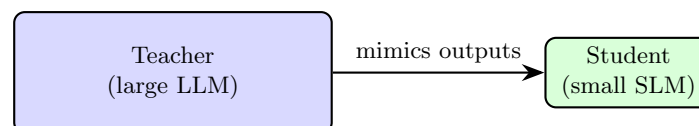


Figure 3: Distillation: a smaller student model is trained to reproduce a larger teacher model’s outputs.

4.8 Quantization

Distillation shrinks the number of parameters. Quantization takes a different approach: keep the same parameters, but store each one more cheaply.

Definition

Quantization reduces the numeric precision used to store a model's weights (for example, from 32-bit to 8-bit or 4-bit numbers), drastically reducing memory use and inference cost.

Example

Worked example: memory savings for a 7-billion-parameter model.

- FP32 (32-bit, 4 bytes/parameter): $7\text{B} \times 4 \text{ bytes} = 28 \text{ GB}$
- FP16 (16-bit, 2 bytes/parameter): $7\text{B} \times 2 \text{ bytes} = 14 \text{ GB}$
- INT8 (8-bit, 1 byte/parameter): $7\text{B} \times 1 \text{ byte} = 7 \text{ GB}$
- INT4 (4-bit, 0.5 bytes/parameter): $7\text{B} \times 0.5 \text{ bytes} = 3.5 \text{ GB}$

Going from FP32 to INT4 shrinks memory usage by $8\times$, with some – usually small – loss in output quality.

Question

Callback: the Kimi K3 mystery from an earlier lecture. We asked how a 2.8-trillion-parameter model could run on 8 GB of RAM. Quantization is half of the answer: Kimi K3 is quantized to 4 bits per weight. The other half – Mixture-of-Experts, which means most of those 2.8 trillion parameters never need to be loaded at all – is covered in a later lecture. With both techniques combined, the puzzle resolves.

4.9 Summary

We covered how a base, pre-trained model becomes a specialized, well-behaved tool: fine-tuning adapts it to a task using labeled examples; RLHF aligns its behavior with human preference using ranked outputs; Chain of Thought improves reasoning by making it explicit and step-by-step; reasoning models take this further by learning how much reasoning a problem needs; multimodal models extend understanding beyond text; and SLMs, distillation, and quantization are three different levers for making models smaller, cheaper, and faster to run. We also closed the loop on the Kimi K3 mystery from an earlier lecture using quantization – with the rest of the answer waiting in a later lecture.

Exercises

1. **Fine-tuning vs. RLHF.** A company wants their support chatbot to (a) know their specific return policy, and (b) always respond in a polite, calm tone even when customers are angry. Which of these goals is better addressed by fine-tuning, and which by RLHF? Justify your answer.
2. **Chain of Thought.** Write a short multi-step word problem of your own, then solve it twice: once as a direct answer, and once using explicit Chain-of-Thought steps.
3. **Reasoning models.** Why might a reasoning model be slower and more expensive to use than a standard chat model for a simple question like “What is the capital of Japan?” Should it be used for every query? Why or why not?
4. **Quantization math.** A model has 13 billion parameters. Compute its memory footprint at FP32, FP16, INT8, and INT4. If you only have 16 GB of RAM available, which of these precisions could you actually run it at?

5. **Distillation vs. quantization.** Both techniques make models cheaper to run, but they work in different ways. Explain the difference in your own words, and describe a situation where you might prefer one over the other.

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